

Functional

- Users can submit a long URL and receive a unique shortened URL.
- Users can access the original URL by visiting the shortened URL.
- The system should redirect users to the original URL with minimal delay.
- Each generated short URL should be unique.
- Users can delete or deactivate an existing short URL .

Non-Functional

- The system should provide **high availability** (e.g., 99.99% uptime).
- The system should support **high scalability** to handle millions of users and billions of URLs.
- URL redirection should have **low latency** (preferably under 100 ms).
- URL creation should be fast and efficient.
- The system should ensure **high reliability**, so every short URL always redirects to the correct destination.

Estimation

Assumption that we are providing and how many users we have :

- Total Registered Users = **10 Million**
- Daily Active Users = **1 Million**
- New Short URLs Created per Day = **10 Million**
- URL Redirects per Day = **100 Million**
- Read : Write Ratio = **10 : 1**
- Average Original URL Length = **200 Bytes**
- Average Short URL Length = **8 Characters**
- Metadata (ID, timestamps, etc.) = **100 Bytes**
- Total Storage per URL ≈ **350 Bytes**

Requirements according to the esti

- ## Traffic
 - Average Write Requests: 116 requests/sec
 - Peak Write Requests: 350 requests/sec
 - Average Read Requests: 1,157 requests/sec
 - Peak Read Requests: 3,500 requests/sec
 - Total Peak Requests: ~3,850 requests/sec
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- ## Storage
 - Storage Per URL: 350 Bytes
 - Daily Storage: 3.5 GB
 - Monthly Storage: 105 GB
 - Yearly Storage: 1.28 TB
 - Storage for 5 Years: 6.4 TB
 - Database Storage (including replicas & indexes): ~20 TB
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- ## Cache
 - Cache Strategy: Store the most frequently accessed URLs
 - Cache Size: 2–4 GB
 - Cache Technology: Redis
- Application Servers: 6–8
- Load Balancers: 2
- Redis Cache Nodes: 2
- Database Servers:
 - 1 Primary Database
 - 2 Read Replicas
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- ## Database Requirements

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- Total Database Storage: ~20 TB
- SSD Storage
- 64–128 GB RAM per Database Server



